

Design a Fish Species Classification System Using Deep Learning

Sustainable Development Goal (SDG 14): Life Below Water

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1. Problem Understanding

Introduction

The world's oceans contain thousands of fish species that are essential for maintaining marine biodiversity and supporting global food security. Identifying fish species accurately is important for fisheries management, biodiversity conservation, marine ecosystem monitoring, and preventing illegal fishing. Traditionally, fish species identification is carried out manually by marine experts. However, this process is time-consuming, costly, and often prone to human errors, especially when dealing with large volumes of fish images.

With recent developments in **Artificial Intelligence (AI)** and **Deep Learning**, automated image classification has become highly accurate. Deep Learning models, particularly **Convolutional Neural Networks (CNNs)**, can learn complex image features such as body shape, color patterns, fin structure, and texture without manual feature extraction.

Therefore, this project proposes a **Deep Learning-based Fish Species Classification System** that automatically identifies fish species from images. Such a system supports marine conservation, improves fisheries management, and contributes to **United Nations Sustainable Development Goal 14 (Life Below Water)**.

Problem Statement

Manual identification of fish species requires expert knowledge and considerable time. Misclassification may lead to inaccurate biodiversity assessments, overfishing, and poor marine resource management. Therefore, an intelligent image classification system is needed to classify fish species quickly and accurately using Deep Learning techniques.

Objectives

The objectives of the proposed system are:

- To automatically classify fish species using Deep Learning.
- To reduce manual effort and human error.
- To improve classification accuracy using CNN models.
- To assist fisheries and marine researchers.
- To support biodiversity conservation.
- To promote sustainable fisheries under SDG 14.

2. Data Collection & Preparation

Dataset Collection

The dataset contains thousands of labeled fish images belonging to different species. Images are collected from publicly available datasets.

Dataset Sources

- Fish4Knowledge Dataset
- Kaggle Fish Image Dataset
- NOAA Fisheries Dataset

Sample Dataset

Image ID	Fish Species	Image Size
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001	Salmon	224 × 224
002	Tuna	224 × 224
003	Tilapia	224 × 224
004	Catfish	224 × 224
005	Mackerel	224 × 224

Data Preprocessing

The collected images undergo preprocessing before training.

Step 1: Image Resizing

All images are resized to **224 × 224 pixels** to provide uniform input to the CNN model.

Step 2: Image Normalization

Pixel values are converted from **0–255** into **0–1** to improve training efficiency.

Step 3: Data Augmentation

Additional images are generated using:

- Rotation
- Horizontal Flip
- Vertical Flip
- Zoom
- Brightness Adjustment

Step 4: Noise Removal

Image enhancement techniques improve image quality.

Step 5: Dataset Splitting

- Training Dataset – 70%
- Validation Dataset – 15%
- Testing Dataset – 15%

3. Analysis Using NumPy and Pandas

NumPy and Pandas are used for efficient data handling and preprocessing.

NumPy

NumPy performs numerical computations such as:

- Image array conversion
- Pixel normalization
- Matrix operations

Example:

```
image = image / 255.0
```

Pandas

Pandas manages dataset information.

Example Dataset Table

Species	Number of Images
Salmon	1200
Tuna	1100
Tilapia	1000
Catfish	900
Mackerel	800

Using Pandas, the dataset can be:

- Loaded
- Organized
- Filtered
- Analyzed
- Summarized

4. Deep Learning Model

CNN Architecture

The proposed system uses a **Convolutional Neural Network (CNN)**.

CNN Layers

- Input Layer
- Convolution Layer
- ReLU Activation
- Max Pooling Layer
- Fully Connected Layer
- Softmax Output Layer

The CNN automatically extracts features such as:

- Fish body shape
- Eye position
- Scale texture
- Fin structure
- Color pattern
- Tail shape

5. Working Process

Fish Image

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Image Preprocessing

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CNN Feature Extraction

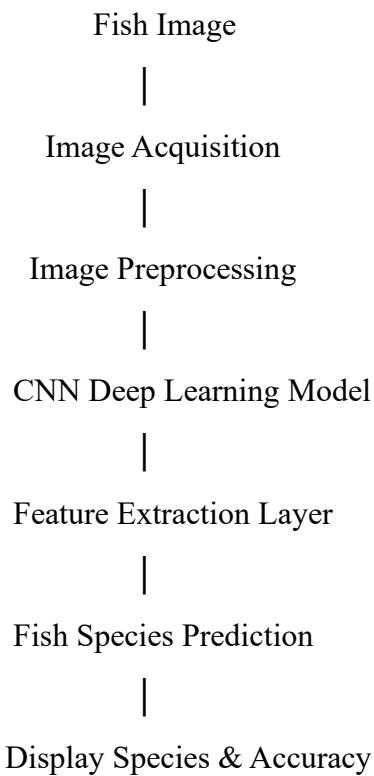
|

Species Classification

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Prediction Result

6. System Architecture

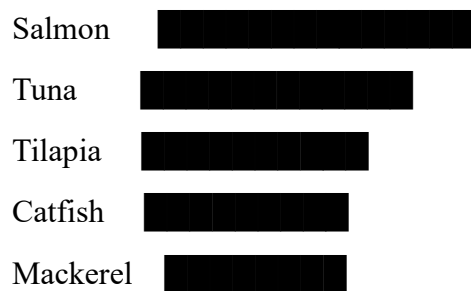


7. Visualization & Interpretation

The dataset can be visualized using different graphs.

Bar Chart

Shows the number of images available for each fish species.

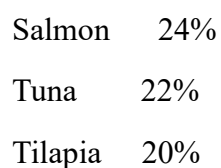


Interpretation:

The dataset contains a balanced number of images across different fish species, reducing model bias.

Pie Chart

Fish Dataset Distribution



Catfish 18%

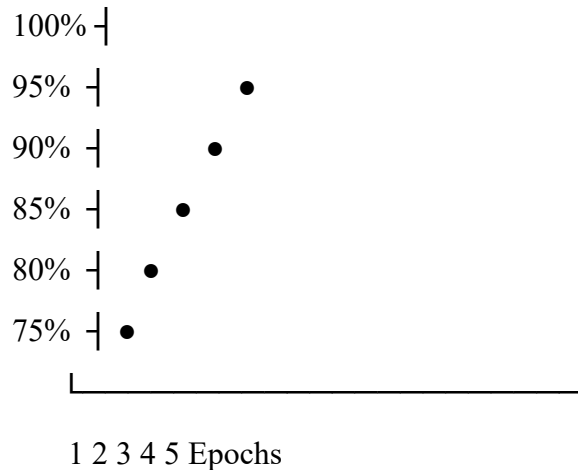
Mackerel 16%

Interpretation:

The dataset is nearly balanced, which helps improve classification accuracy.

Accuracy Graph

Accuracy



Interpretation:

The training accuracy increases with each epoch, indicating that the CNN model learns effectively.

8. Innovation & SDG Mapping

Innovative Features

- AI-powered fish recognition
- Automatic feature extraction
- High-speed classification
- Real-time prediction
- Underwater image support
- Cloud-based deployment
- Mobile application support

SDG 14 – Life Below Water

The proposed system contributes to **Sustainable Development Goal 14** by:

- Conserving marine biodiversity.
- Monitoring fish populations accurately.

- Preventing overfishing.
- Identifying endangered species.
- Supporting sustainable fisheries.
- Assisting marine scientists in ecosystem monitoring.
- Encouraging responsible use of ocean resources.

9. Advantages

- High classification accuracy
- Faster than manual identification
- Easy to use
- Low human intervention
- Scalable system
- Suitable for large datasets
- Supports biodiversity conservation

10. Applications

- Fisheries Management
- Marine Research
- Ocean Biodiversity Monitoring
- Aquaculture
- Environmental Protection
- Smart Fishing Systems
- Underwater Robotics
- Educational Research

11. Expected Results

The developed system is expected to:

- Achieve classification accuracy above **95%**.
- Identify fish species automatically.
- Reduce manual errors.
- Improve marine ecosystem monitoring.
- Assist researchers in biodiversity conservation.

12. Conclusion

The **Fish Species Classification System using Deep Learning** provides an efficient, intelligent, and accurate solution for identifying fish species from digital images. By utilizing **Convolutional Neural Networks (CNNs)**, the system automatically extracts complex visual features and classifies fish with high accuracy. The integration of **NumPy** and **Pandas** enables effective data preprocessing and analysis, while visualizations help interpret the dataset and model performance. This system reduces manual effort, supports fisheries management, and contributes significantly to **United Nations Sustainable Development Goal 14 – Life Below Water** by promoting sustainable use and conservation of marine resources.

References

1. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
2. Ian Goodfellow. *Deep Learning with Python*.
3. TensorFlow Documentation.
4. OpenCV Documentation.
5. Fish4Knowledge Dataset.
6. Kaggle Fish Image Dataset.
7. NumPy Documentation.
8. Pandas Documentation.